

ESSE ASSIGNMENT TECHNICAL REPORT

ENERGY-INEFFICIENT BUILDINGS

**Green Building Energy Management
Portal**

**Neethu KP
S3 CS C
Roll no.71
U2503287**

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ABSTRACT

Energy-efficient buildings are important for reducing electricity consumption, operating costs and the environmental impacts associated with energy generation. Buildings consume electricity for lighting, air conditioning, fans, computers, laboratory equipment, lifts, pumps and other services. Energy can be wasted when equipment is operated unnecessarily, when buildings are occupied inefficiently, or when users do not have clear information about their consumption. This report proposes **GreenPulse**, a web-based Green Building Energy Management Portal designed to provide a centralized platform for monitoring and improving building energy performance.

The proposed system allows users and authorized facility managers to record or import electricity readings, view energy consumption by building or room, compare present consumption with historical values, identify unusual consumption patterns and receive practical energy-saving recommendations. A dashboard provides visual information about daily, weekly and monthly energy use. The system also introduces an Energy Action Score that converts consumption data into simple actions for users. An administrative interface can be used to manage buildings, zones, readings, alerts and sustainability reports.

The proposed solution primarily supports **SDG 7** – Affordable and Clean Energy and also contributes to **SDG 11** – Sustainable Cities and Communities, **SDG 12** – Responsible Consumption and Production and **SDG 13** – Climate Action. The system can support environmental management practices associated with ISO 14001 by providing measurable information, monitoring performance and supporting continual improvement. A basic life-cycle perspective is also considered because the environmental impact of the digital platform includes the energy and resources required for its development, hosting and operation.

GreenPulse can improve energy awareness, identify avoidable electricity consumption, support faster corrective action and provide useful historical information for building-management decisions. The solution requires relatively low infrastructure for a basic prototype and can later be expanded using smart meters, IoT sensors and intelligent energy analytics.

INTRODUCTION

Energy is one of the most important resources required for the operation of modern buildings. Educational institutions, offices, hospitals, commercial buildings and residential buildings depend on electricity for lighting, cooling, computing, communication and many other activities. As the number of electrical devices and services increases, inefficient energy use can become a major environmental and economic concern.

Energy inefficiency occurs when a building consumes more energy than necessary to provide its required services. Common causes include lights and fans operating in unoccupied rooms, air conditioners being used unnecessarily, computers and laboratory equipment remaining powered on, inefficient appliances, poor maintenance and the absence of effective monitoring. These activities may appear small individually, but repeated energy waste across many rooms and buildings can result in significant consumption.

Energy efficiency is also connected with climate action. When electricity is generated using fossil fuels, unnecessary electricity consumption can lead to additional greenhouse-gas emissions. Improving building energy efficiency therefore provides an opportunity to reduce resource use while maintaining the required comfort and functionality of the building.

Many buildings monitor energy mainly through monthly electricity bills. Although bills provide total consumption, they may not show which room, floor, equipment category or operating period caused an increase. Continuous or more detailed monitoring can make energy use easier to understand. A digital platform can convert raw energy readings into charts, comparisons, alerts and recommendations that help users take action.

Therefore, technology can be used to bridge the gap between building energy data and practical energy conservation. The proposed GreenPulse portal is designed from an Engineering/IT perspective to provide this connection.

PROBLEM STATEMENT

Existing Problem

Energy-inefficient buildings consume unnecessary electricity because energy use is not always monitored at a useful level of detail. Examples of inefficient practices include:

- Lights, fans and other equipment operating in unoccupied rooms.
- Air conditioners operating outside required schedules.
- Computers and laboratory equipment remaining powered on or in standby mode.
- Use of old or inefficient electrical appliances.
- Poor scheduling of energy-intensive equipment.
- Lack of room-wise or floor-wise energy information.
- Delayed identification of abnormal electricity consumption.

A major problem is that building users may know that energy should be saved but may not know where the largest avoidable consumption occurs. Facility managers may receive total electricity information without having a simple way to identify the locations and periods responsible for unusual consumption. Another issue is that energy data, maintenance information and user actions may be stored separately.

Existing limitations

- Lack of centralized building-energy monitoring.
- Limited room-wise or floor-wise consumption information.
- Difficulty identifying abnormal consumption quickly.
- Limited visibility of energy-saving actions and their results.
- Lack of simple feedback for ordinary building users.

Therefore, the proposed system should not simply display an electricity value. It should combine energy monitoring, comparison, alerts, recommendations and sustainability analytics into one system.

SUSTAINABILITY ANALYSIS

1. Relevant Sustainable Development Goals

SDG 7 – Affordable and Clean Energy

The primary SDG associated with this project is SDG 7. Sustainable energy use requires both access to energy and efficient consumption. GreenPulse supports this objective by helping buildings monitor electricity use, identify waste and encourage efficient operation. The platform can also be extended to display renewable-energy generation from sources such as rooftop solar systems.

SDG 11 – Sustainable Cities and Communities

Buildings form an important part of cities and communities. Improving their energy performance can reduce resource consumption and contribute to more sustainable infrastructure. The proposed portal provides information that can help institutions manage energy use more effectively.

SDG 12 – Responsible Consumption and Production

The system promotes responsible consumption by making electricity use visible to users and facility managers. It can encourage switching off idle equipment, improving operating schedules and identifying inefficient devices. Historical data can also support more responsible maintenance and replacement decisions.

SDG 13 – Climate Action

Reducing unnecessary electricity consumption can contribute to lower greenhouse-gas emissions when the consumed electricity is associated with carbon-intensive generation. GreenPulse can display estimated emissions reductions using a clearly stated electricity emission factor.

2. ISO 14001 Analysis

ISO 14001 provides requirements and guidance for environmental management systems. GreenPulse can support an ISO 14001-oriented environmental management approach by providing information for planning, monitoring, documentation and continual improvement.

ISO 14001-Oriented Application

Environmental Policy: The organization can establish objectives for energy efficiency, conservation and continual environmental improvement.

Planning: Energy consumption data can help identify significant energy-use areas and environmental risks associated with inefficient operation.

Implementation: The platform provides a mechanism for communicating energy-saving recommendations and assigning corrective actions.

Monitoring: The dashboard can monitor electricity consumption, energy trends, abnormal-use alerts, energy-saving actions and improvement over time.

Continual Improvement: Historical data can be reviewed to identify recurring energy waste and improve building-management practices.

PROPOSED ENGINEERING/IT SOLUTION

1. GreenPulse System

GreenPulse – Green Building Energy Management Portal is a responsive web application that helps building users and facility managers monitor electricity consumption and improve energy efficiency. The system has two main interfaces.

User Interface: Users can view energy information, check consumption trends, compare zones, view energy-saving recommendations and understand the Energy Action Score.

Management Interface: Authorized facility managers can add or import readings, manage buildings and zones, view alerts, monitor improvement and generate sustainability reports.

2. Energy Monitoring Categories

- Lighting
- Air Conditioning and Ventilation
- Computers and IT Equipment
- Laboratory and Office Equipment
- Fans and Other Electrical Appliances
- Common-Area Equipment

PROPOSED ENGINEERING/IT SOLUTION

3. Energy Monitoring Process

The basic workflow of the proposed system is:

- Energy reading collected or entered → Building/zone selected → Data validated and stored.
- Current reading compared with historical or baseline consumption.
- System identifies normal or unusual consumption patterns.
- Alert is generated when configurable conditions indicate possible inefficiency.
- Recommended action is displayed to the responsible user or facility manager.
- Later readings are compared to evaluate whether consumption improved.

This process maintains a digital record of energy performance and connects monitoring with corrective action.

4. Unique Features of GreenPulse

- Room/Floor Energy Monitoring: Allows consumption to be compared across different building zones.
- Energy Action Score: Converts energy data into a simple score indicating whether attention is required.
- Smart Consumption Alerts: Identifies unusual increases or consumption outside expected operating periods.
- Action Recommendations: Provides practical suggestions based on the detected pattern.
- Energy Trend Dashboard: Displays daily, weekly and monthly consumption using charts.
- Baseline Comparison: Compares present consumption with selected historical values.
- Sustainability Analytics: Shows energy-saving trends and estimated environmental benefits.

5. Basic Technical Architecture

The website can use HTML, CSS and JavaScript for the front end, with Java, Python or another suitable backend technology for application logic. MySQL or PostgreSQL can be used to store user, building, zone, equipment, reading and alert information. Charting libraries can be used for visualizing consumption

trends. The prototype can initially use manually entered readings and later integrate smart meters or IoT devices through APIs.

FEASIBILITY ANALYSIS

1. Technical Feasibility

GreenPulse is technically feasible because it can be developed using commonly available technologies such as HTML, CSS, JavaScript, Java or Python, MySQL, charting libraries and cloud hosting. The basic prototype does not require specialized hardware because readings can initially be entered manually. Smart-meter and IoT integration can be added later.

2. Economic Feasibility

The prototype can be developed using open-source software and student-level development tools, keeping initial costs low. The main future expenses would include hosting, cloud storage, database services, smart meters, sensors, networking and maintenance. Energy savings generated by the system can later be compared with deployment costs.

3. Operational Feasibility

GreenPulse is designed to be simple to use. Users can view their building or room information, check energy trends and follow recommendations. Facility managers can add readings, review alerts and monitor improvement through a dashboard. A simple interface reduces the training required.

4. Social Feasibility

The system encourages students, staff and facility managers to participate in energy conservation. By making consumption visible and providing understandable recommendations, the platform can increase awareness and encourage responsible energy use.

5. Legal and Ethical Feasibility

The system must protect user and building data. Important measures include secure authentication, access control, password protection, appropriate data retention and limiting access to administrative information. Occupancy-related information should not be collected or displayed unnecessarily.

EXPECTED ENVIRONMENTAL BENEFITS

1. Reduced Electricity Waste – Helps identify unnecessary electricity use from idle equipment, lighting and poor operating schedules.

2. Better Energy Monitoring – Provides continuous or periodic information about building energy performance.

3. Identification of Inefficient Zones – Helps identify rooms, floors or areas with unusually high consumption.

4. Faster Corrective Action – Alerts and recommendations can help users respond to energy problems sooner.

5. Improved Equipment Management – Historical information can support maintenance and replacement decisions.

6. Increased Energy Awareness – Users can see how their daily practices influence building energy consumption.

7. Support for Renewable Energy – Future versions can integrate solar generation and compare renewable production with building demand.

The actual amount of energy or carbon reduction cannot be guaranteed before real building data is collected. The recommended approach is to establish a baseline and compare later consumption with the baseline after energy-saving actions are implemented.

Suggested Performance Indicators

- Daily, weekly and monthly electricity consumption.
- Energy consumption by room, floor or building.
- Consumption during occupied and unoccupied periods.
- Number of abnormal-consumption alerts.
- Number of alerts resolved.
- Energy consumption before and after corrective actions.
- Estimated energy and carbon savings based on documented assumptions.

JUSTIFICATION OF THE PROPOSED SOLUTION

GreenPulse is justified because building energy management is not only a hardware problem but also an information and decision-making problem. Simply asking users to save electricity may not produce measurable improvement because users cannot easily see where energy is being wasted. On the other hand, replacing every electrical device immediately may be expensive. A web-based monitoring system provides a practical intermediate solution by making existing energy information useful.

The proposed platform improves on a basic energy dashboard by integrating monitoring, baseline comparison, abnormal-use alerts, an Energy Action Score and practical recommendations. This makes the system a digital energy-management and decision-support platform rather than simply a website that displays electricity readings.

The project also has strong interdisciplinary value. From a Computer Science perspective, it involves responsive web design, database management, data visualization, user authentication and data analysis. From an engineering perspective, it relates to electrical energy consumption and building operation. From an environmental perspective, it supports energy conservation and climate action.

The system is scalable. A small institution can initially monitor one building using manually entered readings. Later, it can be expanded to multiple buildings and connected to smart meters, IoT sensors, occupancy information and renewable-energy systems. Historical data can also be used for predictive analytics and anomaly detection.

Therefore, GreenPulse provides a practical, low-infrastructure starting point while maintaining the possibility of future intelligent energy management.

CONCLUSION

Energy-inefficient buildings are an important environmental and operational problem because unnecessary electricity consumption increases resource use, operating costs and the environmental impacts associated with electricity generation. Effective energy management requires not only awareness but also measurement, monitoring and timely action.

This report proposed GreenPulse – Green Building Energy Management Portal, a web-based platform that integrates energy monitoring, room and floor comparison, baseline analysis, abnormal-consumption alerts, an Energy Action Score and practical recommendations. The system is intended to create a structured connection between building energy data and the actions required to reduce avoidable consumption.

The proposed system primarily supports SDG 7 – Affordable and Clean Energy, while also contributing to SDGs 11, 12 and 13. It can support environmental-management practices associated with ISO 14001 by improving monitoring, documentation and continual improvement. The project is technically, economically and operationally feasible because a basic prototype can be developed using commonly available web technologies and can operate initially without specialized hardware.

The major potential benefit of GreenPulse is that it can make energy consumption visible and actionable. By gradually integrating real meter data, smart devices and intelligent analytics, the system can evolve from a student-level website into a practical building energy-management platform.

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